Experiment No. 6

Aim: Design an app with location tracking using Google Maps API.

Apparatus / Software Required :

Hardware	- Laptop(ASUS Vivobook 16x) - Minimum 8 GB RAM - AMD Ryzen 5 5600H with Radeon Graphics Processor - Internet Connection
Software	- Windows 11 Operating System - Android Studio (Quail) - Java Development Kit (JDK) - Android SDK - Android Emulator (Pixel 7) - Gradle Build System

Theory

The Google Maps SDK for Android lets an app embed an interactive map using a SupportMapFragment. The map is delivered asynchronously through the OnMapReadyCallback interface, which supplies a GoogleMap object used to add markers, polylines and to move the camera. Every request is authenticated with an API key stored in the manifest as the meta-data com.google.android.geo.API_KEY.

Location is obtained through the FusedLocationProviderClient of Google Play services, which combines GPS, Wi-Fi and cell data to give an accurate and battery-efficient position. A LocationRequest describes how often and how accurately updates are required and a LocationCallback receives each Location. Because location is dangerous data, ACCESS_FINE_LOCATION / ACCESS_COARSE_LOCATION must be declared in the manifest and also requested from the user at run time (Android 6.0+).

Tracking is achieved by storing every received coordinate as a LatLng, drawing them as a Polyline and summing Location.distanceTo() between consecutive fixes to obtain the distance travelled.

Implementation Steps:

1. Project Creation: Create an Empty Views Activity in Android Studio named Map Tracker, using Java and Minimum SDK API 24.

2. API Setup: Enable Maps SDK for Android in Google Cloud Console, generate an API key, restrict it, and configure billing.

3. UI Design: Create a layout with Google Maps, latitude/longitude, distance display, and Start, Stop, My Location, and Clear Path buttons.

4. Permissions: Add Internet and location permissions in AndroidManifest.xml and request runtime location permission.

5. Implementation: Initialize Google Maps and use the Fused Location Provider to track location every 3 seconds or after 2 metres of movement.

6. Functionality: Display the current location with a marker, draw the movement path using a polyline, calculate distance, and implement Start/Stop Tracking and Clear Path.

7. Testing: Run the app on a device or emulator, grant location permission, start tracking, and verify the location marker, path, and distance updates.

Files Used / Code Changes

File	Purpose / Change
MainActivity.java	Map setup, permission handling, location updates, polyline drawing and distance calculation.
activity_main.xml	Layout: map fragment, information TextView and four buttons.
AndroidManifest.xml	Permissions, Maps API-key meta-data, launcher activity.
google_maps_api.xml	Holds the API key string (placeholder only - add your own key).
strings.xml / colors.xml / themes.xml	Text, colours and the Material theme.
app/build.gradle	SDK versions, Java compatibility and Maps/Location dependencies.
settings.gradle, build.gradle, gradle.properties	Project-level Gradle configuration (AndroidX enabled).
ic_launcher.xml	Vector drawable used as the app icon.

Components Used

Component	Use in this project
AppCompatActivity	Host activity (MainActivity).
SupportMapFragment / GoogleMap	Displays the map and controls markers, polyline and camera.
FusedLocationProviderClient	Provides location updates and last known location.
LocationRequest / LocationCallback	Defines update rate/accuracy and receives locations.
Marker, Polyline	Show current position and travelled path.
TextView, Button, LinearLayout	Information display, controls and layout.
Toast	Short user messages (permission denied, no location).

Important XML Attributes Used

Attribute	Meaning
android:name="com.google.android.gms.maps.SupportMapFragment"	Class of the fragment that hosts the map.
meta-data android:name="com.google.android.geo.API_KEY"	Supplies the Maps API key to the SDK.
uses-permission ACCESS_FINE_LOCATION / COARSE_LOCATION	Allow access to precise / approximate location.
uses-permission INTERNET	Needed to download map tiles.
android:layout_weight="1" with layout_height="0dp"	Makes the map fill the remaining screen height.
android:exported="true"	Launcher activity must be exported on Android 12+.
android:enabled="false"	Stop button disabled until tracking starts.

Important Java Methods Used

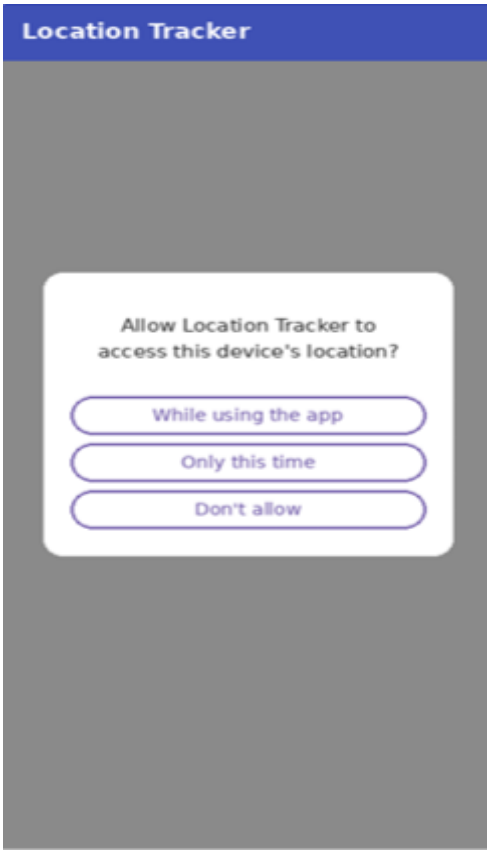
Method / Class	Purpose
getMapAsync(OnMapReadyCallback)	Requests the GoogleMap object asynchronously.
onMapReady(GoogleMap)	Callback in which map settings and camera are initialised.
registerForActivityResult(...)	Runtime permission request and result handling.
LocationServices.getFusedLocationProviderClient()	Creates the location client.
requestLocationUpdates() / removeLocationUpdates()	Start / stop receiving location updates.
getLastLocation()	Gets the most recent cached location.
onLocationResult(LocationResult)	Receives new locations.
addPolyline(), addMarker(), Polyline.setPoints(), Marker.setPosition()	Draw and update path and marker.
animateCamera(CameraUpdateFactory.newLatLngZoom())	Moves the camera smoothly to the current position.
Location.distanceTo()	Distance in metres between two locations.
setMyLocationEnabled(true)	Shows the blue My Location dot.

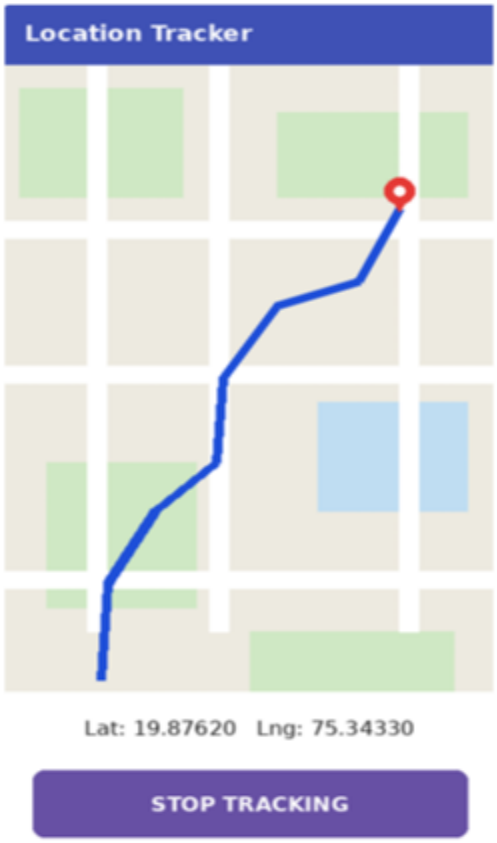
Errors Faced & Troubleshooting

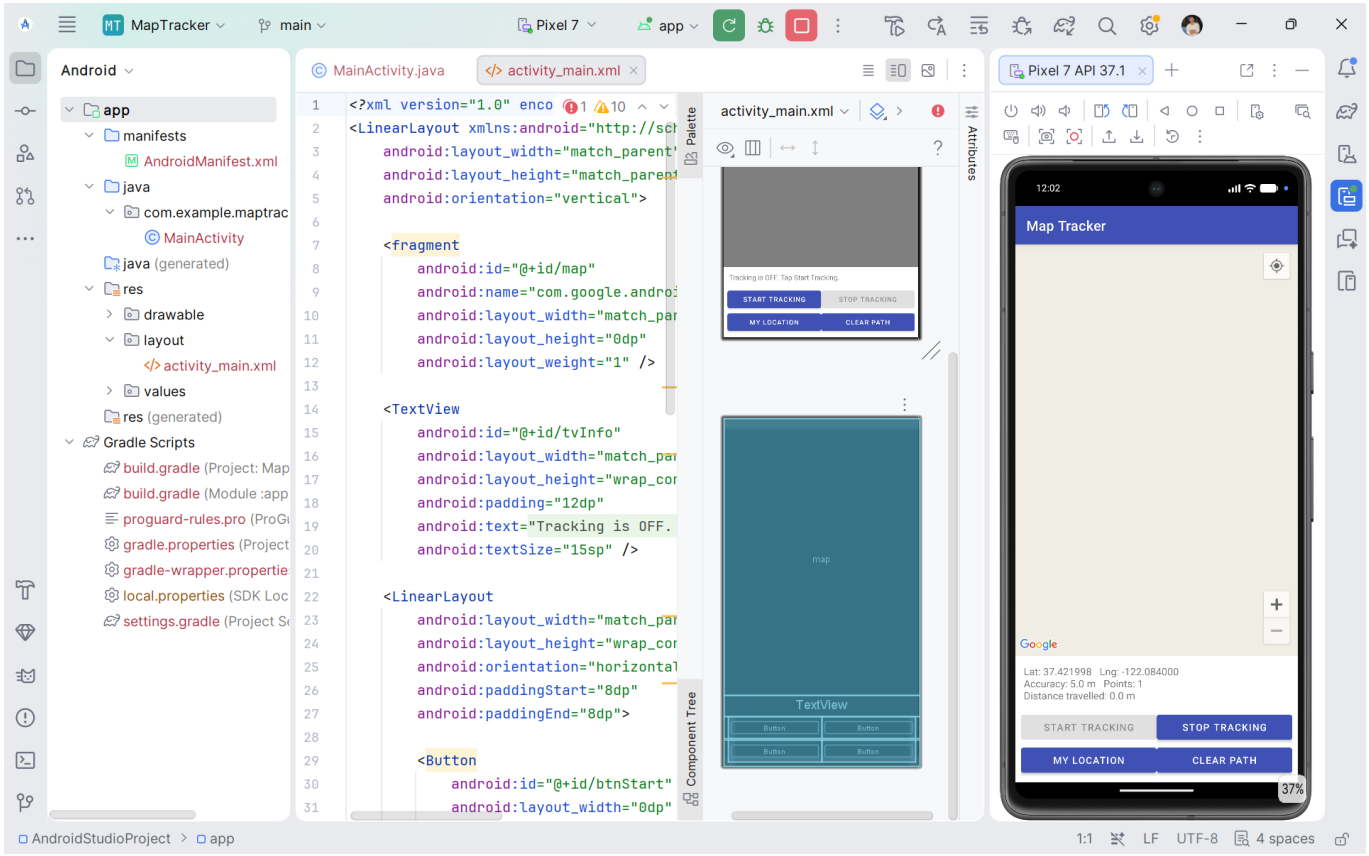
The following are common errors for this experiment and their fixes. Add any additional errors you personally encounter while running it.

Error / Problem	Cause and Fix
Blank grey map / 'Authorization failure' in Logcat	Wrong or missing API key, Maps SDK for Android not enabled, billing not linked, or key restriction (package/SHA-1) does not match. Fix the key and restrictions.
Gradle sync failed: could not resolve play-services	No internet or google() repository missing. Check connection and settings.gradle repositories.
Location never updates on emulator	No mock location set. Use Extended controls > Location and press Send, or run a route.
Permission denied toast	User denied permission. Enable it in Settings > Apps > Map Tracker > Permissions.
App crashes: 'API key not found'	meta-data tag missing or placed outside the <application> element.
Marker/path not drawn indoors on a phone	GPS signal weak. Go outdoors and enable high-accuracy location mode.

Output







Conclusion

The experiment demonstrates how to integrate the Google Maps SDK and the Fused Location Provider into an Android application, request location permissions at run time, show the live position on a map, draw the travelled path and calculate the distance. The source code is complete for this purpose; it must be built with your own Maps API key and run on a device or emulator to obtain the actual output screenshots.

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